

MIDDLESBORO-BELL COUNTY AP, KY, USA

WMO: 720353

Lat: 36.611N

Lon: 83.738W

Elev: 352

StdP: 97.17

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 63875

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.5	-17.4	0.8	-6.9	-14.3	1.1	-4.4	7.7	8.3	7.0	6.7	1.3	340	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.8	24.4	32.6	23.6	31.6	23.1	26.4	30.8	25.3	30.3	24.5	29.3	1.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.3	28.4	23.9	19.6	27.5	22.9	18.5	26.3	84.4	31.3	79.7	29.9	75.7	29.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
6.3	5.3	4.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-13.3	35.1	3.4	2.5	-15.7	37.0	-17.7	38.5	-19.6	39.9	-22.1	41.7
			-14.9	26.4	4.1	1.5	-17.9	27.5	-20.3	28.3	-22.6	29.1	-25.5	30.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	3.0	4.8	9.4	14.8	19.4	23.4	24.8	24.6	21.6	15.0	8.7	5.1
	DBStd	9.01	6.06	5.92	5.60	4.70	3.94	2.63	2.52	2.41	3.15	4.74	4.72	5.07
	HDD10.0	746	229	164	81	13	1	0	0	0	0	11	78	168
	HDD18.3	2134	477	379	281	124	38	1	0	0	9	124	290	411
	CDD10.0	2417	11	19	62	156	291	402	459	451	348	165	38	16
	CDD18.3	766	0	0	3	17	69	153	201	193	108	21	1	0
	CDH23.3	6799	0	2	45	246	677	1398	1732	1628	869	196	7	0
	CDH26.7	2608	0	0	7	51	213	558	730	681	327	41	0	0
Wind	WSAvg	1.2	1.6	1.7	1.7	1.7	1.1	0.9	0.8	0.7	0.7	0.9	1.1	1.3
Precipitation	PrecAvg	1301	107	102	135	107	125	109	131	108	79	82	106	113
	PrecMax	1694	262	180	311	244	250	230	323	283	146	158	278	233
	PrecMin	1009	31	13	38	12	46	26	28	33	12	0	25	9
	PrecStd	182	54	43	66	48	52	49	73	56	36	43	53	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	22.5	27.6	31.0	32.5	34.2	36.2	35.2	33.1	29.9	24.2	20.7
		MCWB	15.6	16.5	17.7	18.1	22.6	23.8	25.8	25.5	22.3	21.5	16.8	16.7
	2%	DB	17.2	19.8	24.0	27.9	31.0	32.9	33.7	33.5	32.2	27.7	21.9	18.0
		MCWB	13.0	15.4	15.9	17.6	21.3	23.4	25.1	24.6	22.2	19.6	15.2	14.7
	5%	DB	14.9	17.4	22.0	26.3	29.0	32.0	32.3	32.3	30.5	25.9	19.2	16.2
		MCWB	11.8	12.1	14.7	16.5	20.4	23.1	24.3	23.9	22.2	19.0	14.0	13.7
	10%	DB	12.5	14.7	19.0	24.0	27.5	30.0	30.9	30.8	28.6	22.9	17.5	13.7
		MCWB	9.9	10.5	13.2	15.7	19.7	22.2	23.7	23.6	21.5	17.6	13.2	11.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.8	19.4	21.0	24.0	27.2	27.7	28.3	25.5	23.3	18.5	17.7
		MCDB	18.4	20.8	24.6	26.9	30.4	32.2	32.4	32.1	30.3	27.8	21.8	19.7
	2%	WB	14.1	16.0	17.6	19.2	22.7	25.3	26.5	26.2	24.2	21.6	17.1	15.6
		MCDB	16.4	19.2	21.6	24.7	28.2	30.0	31.4	29.9	28.7	25.2	19.8	17.6
	5%	WB	12.4	13.7	15.9	18.0	21.7	24.0	25.5	25.1	23.3	20.0	15.6	13.9
		MCDB	14.0	15.8	20.5	23.4	26.9	29.1	30.6	30.0	27.8	23.4	18.1	15.5
	10%	WB	10.5	11.5	14.3	16.9	20.7	23.0	24.6	24.1	22.4	18.7	13.8	11.8
		MCDB	12.1	14.1	18.2	22.3	25.5	27.8	29.1	29.0	26.5	22.1	16.8	13.4
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.8	12.5	13.8	12.8	12.3	11.2	11.5	12.1	12.9	12.9	9.7
		MCDBR	12.0	14.0	16.3	17.2	15.2	14.5	12.8	13.1	14.5	15.2	15.7	11.7
		MCWBR	8.8	9.1	8.8	7.9	6.5	6.3	5.4	5.7	6.2	7.5	9.4	8.6
	5% WB	MCDBR	9.5	11.2	13.5	13.4	13.0	12.2	11.3	11.7	12.2	12.3	11.3	10.6
		MCWBR	7.9	8.6	7.9	7.1	6.0	5.8	5.1	5.6	5.7	7.1	7.8	8.4
Clear-Sky Solar Irradiance	taub	0.307	0.316	0.347	0.391	0.440	0.467	0.497	0.505	0.405	0.364	0.324	0.305	
	taud	2.506	2.474	2.422	2.302	2.181	2.140	2.089	2.046	2.311	2.421	2.501	2.530	
	Ebn at Noon	891	923	919	891	848	822	794	777	846	854	861	869	
	Edh at Noon	80	93	108	129	147	154	161	164	119	98	80	74	
All-Sky Solar Radiation	RadAvg	2.12	2.71	3.88	5.08	5.64	6.13	5.71	5.42	4.59	3.56	2.57	1.86	
	RadStd	0.20	0.36	0.35	0.35	0.45	0.36	0.33	0.28	0.43	0.46	0.29	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+81

Nomenclature: See separate page